Term 3 2026

FINAL EXAM
ETL PROJECT

Business Intelligence
MADSC301-MUC07611

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Student: **Sujin Lee**

OUR CONTENTS

01. INTRODUCTION

Analyze short-term trends in major U.S. semiconductor stocks

The project uses the **Twelve Data API, Apache Airflow, PostgreSQL**, and **Streamlit** to create an end-to-end workflow for semiconductor stock analysis.

The main objective of this project is to monitor and analyze short-term trends in major U.S. semiconductor stocks through an **automated ETL pipeline and interactive dashboard**. This project follows the assignment requirement to build a complete ETL pipeline including data collection, cleaning, storage, orchestration, and visualization.

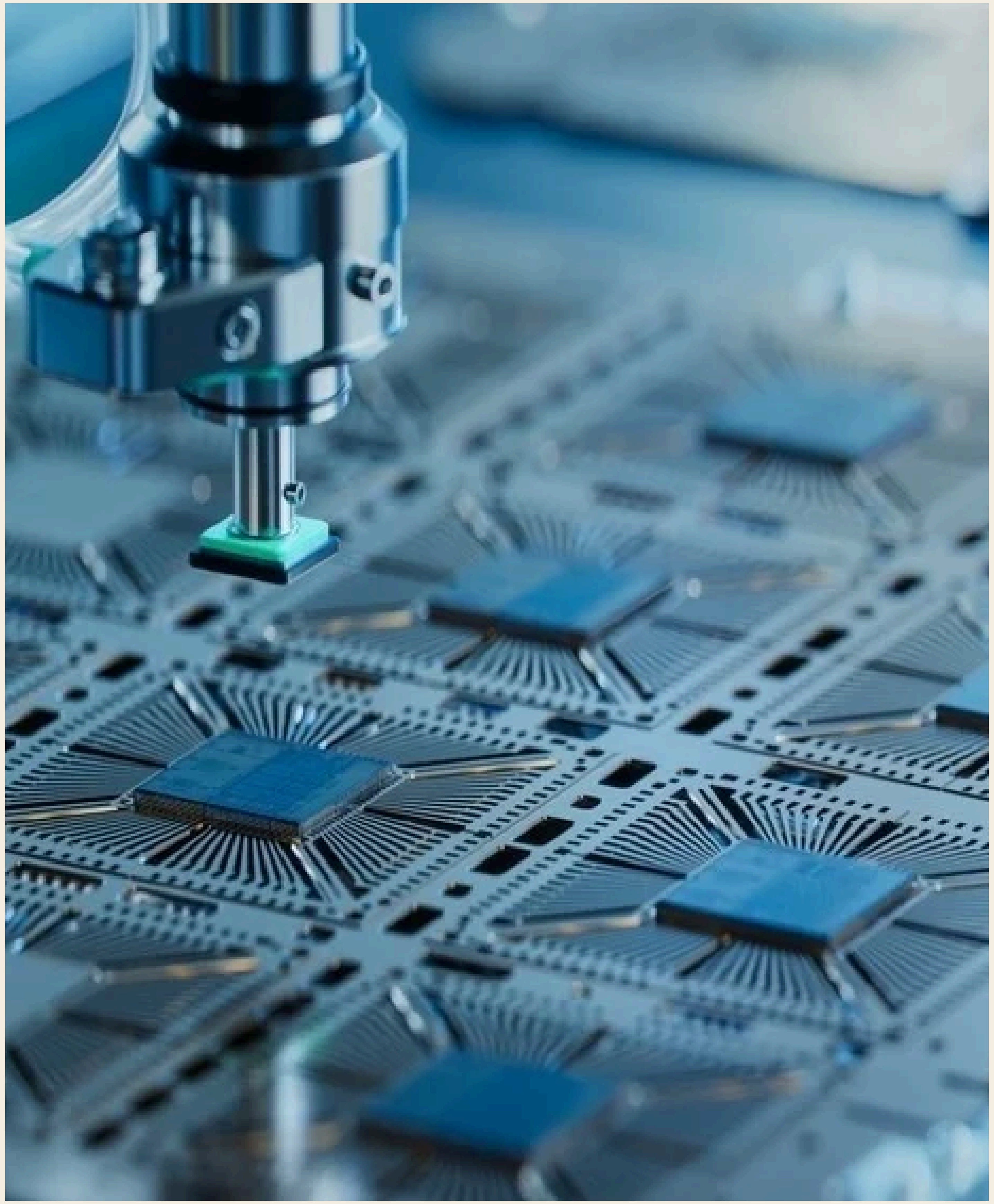
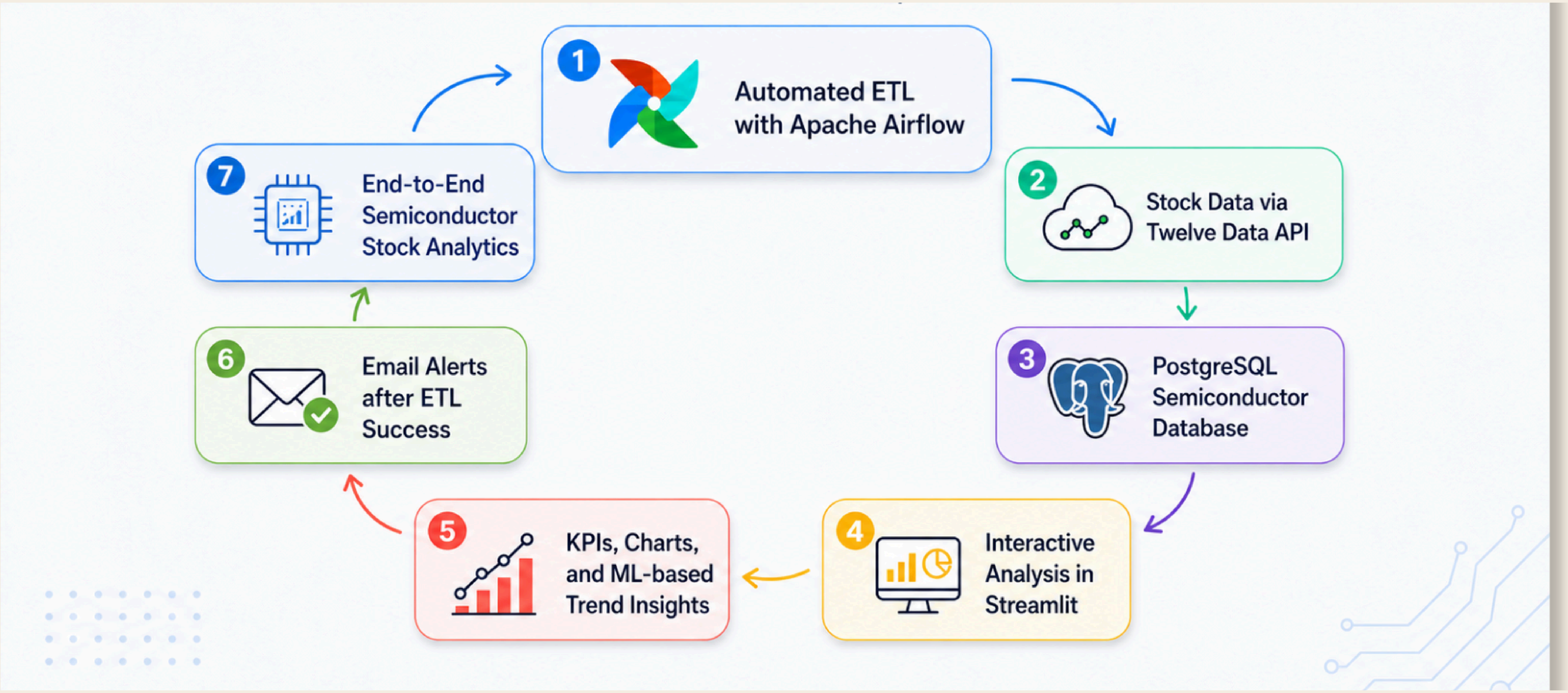
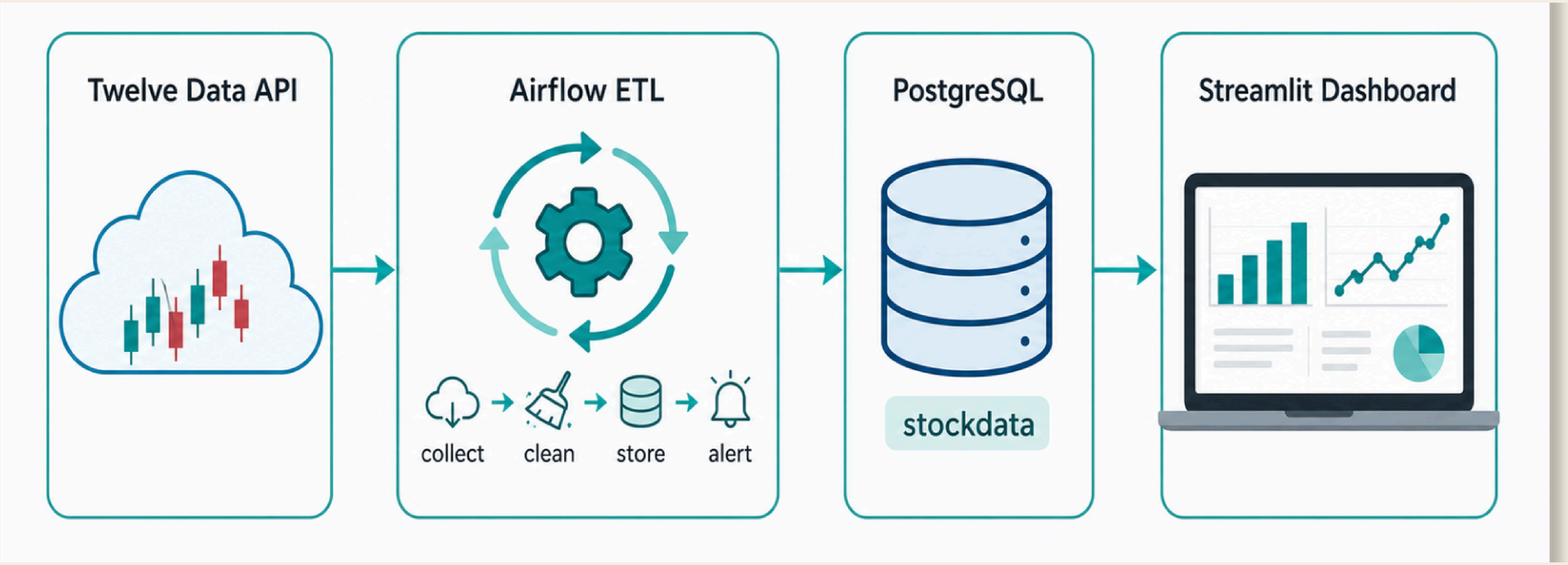


image source

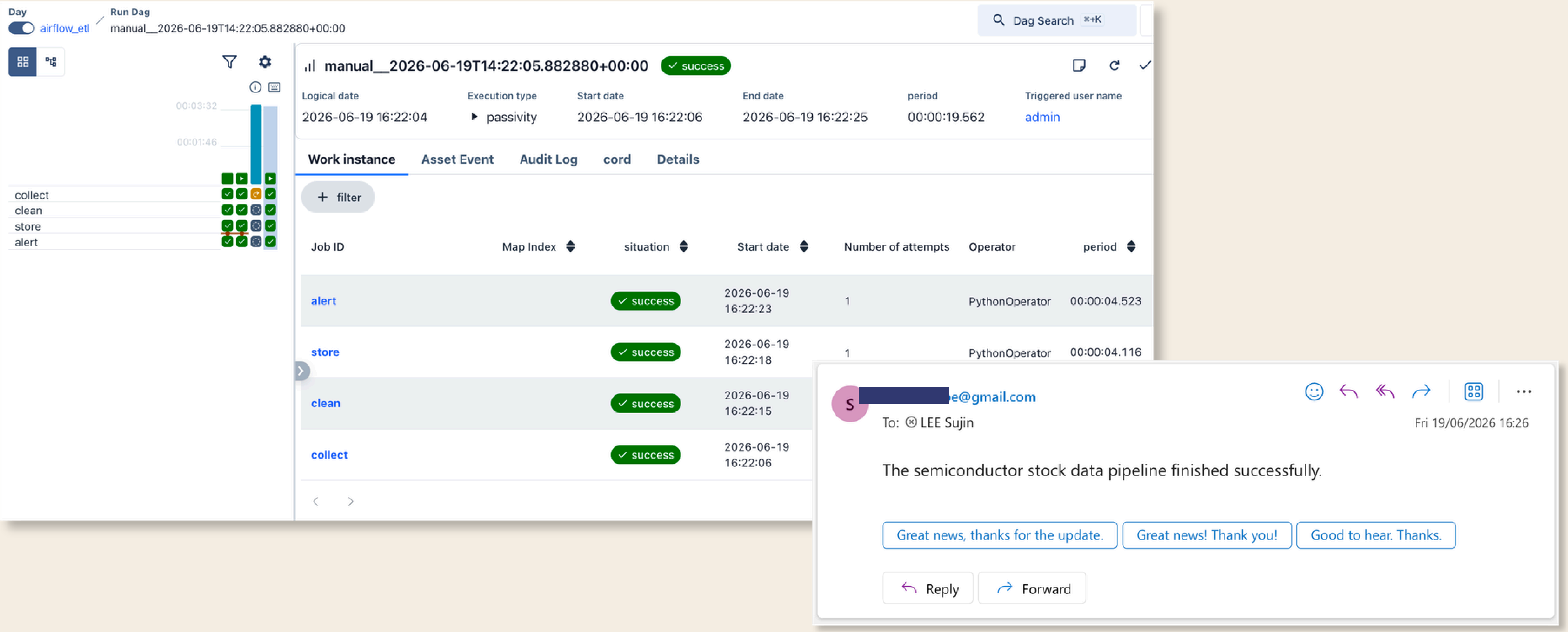
02. KEY FEATURES



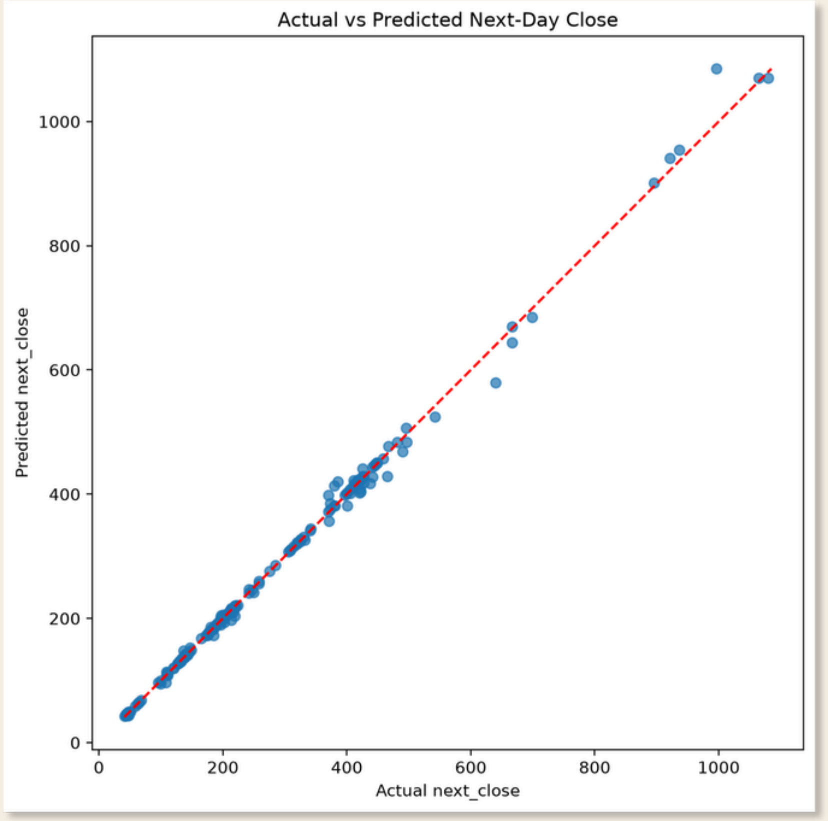
03. ETL ARCHITECTURE



04. ARIFLOW AUTOMATION

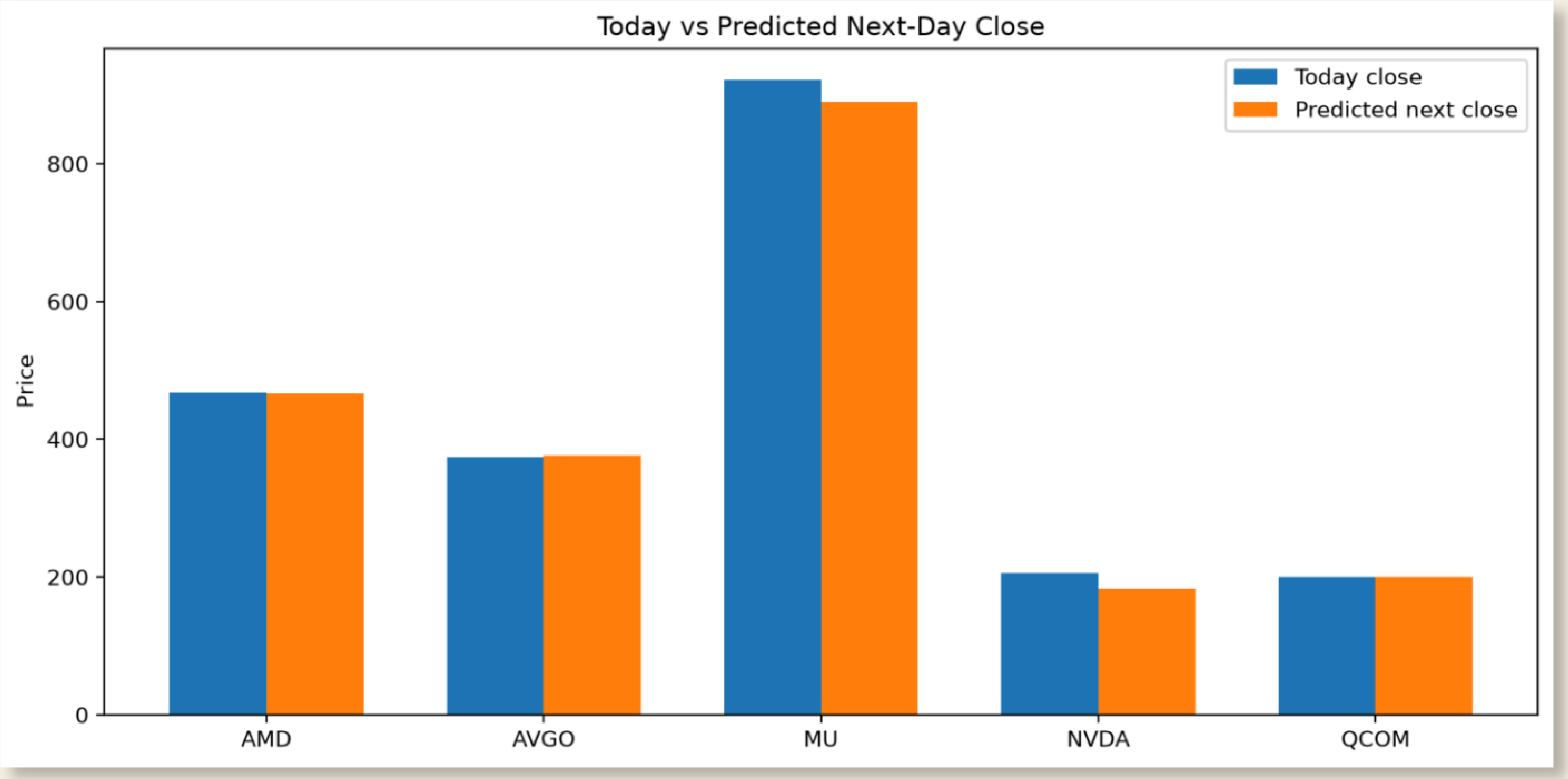


05. MACHINE LEARNING

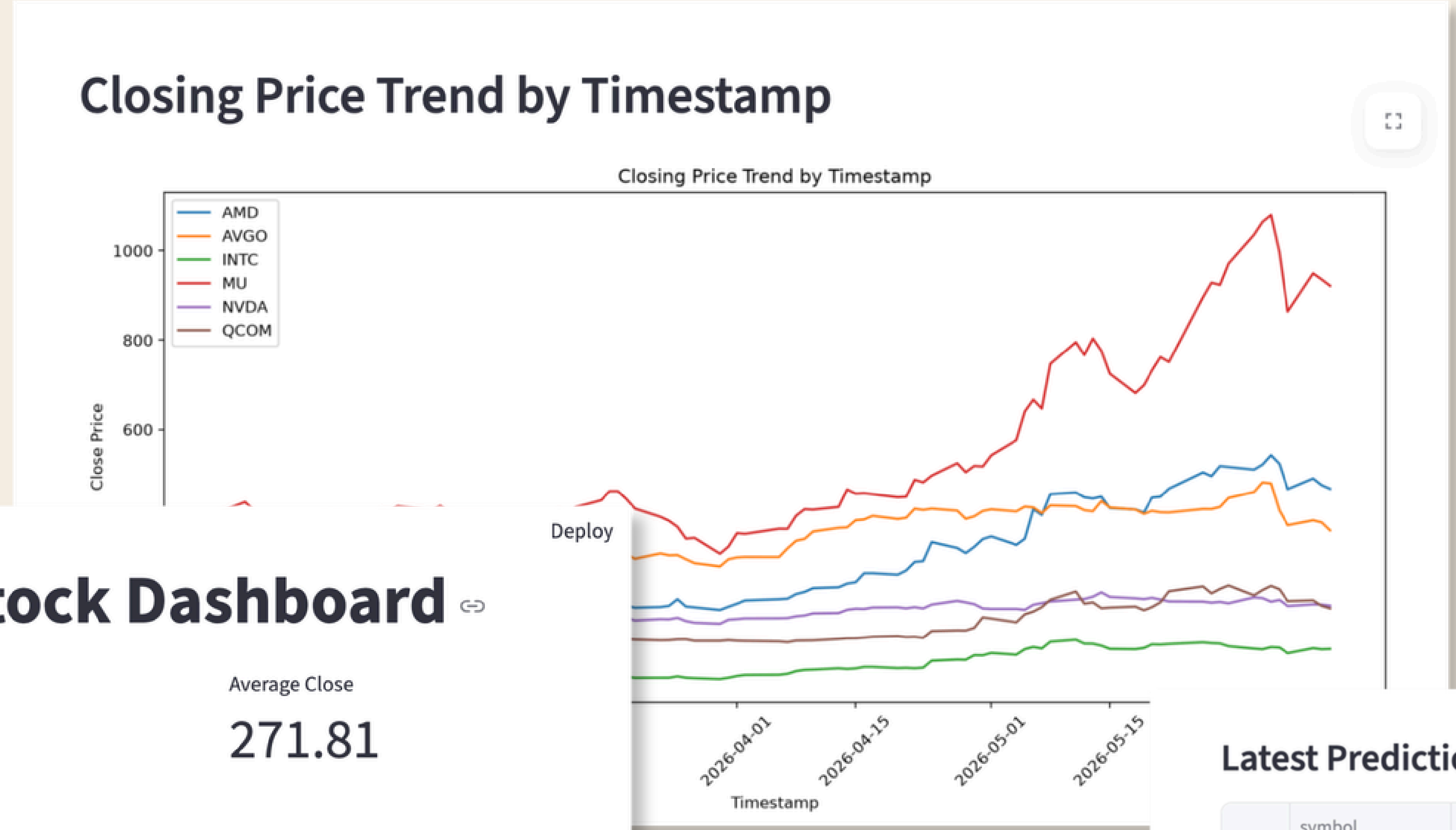


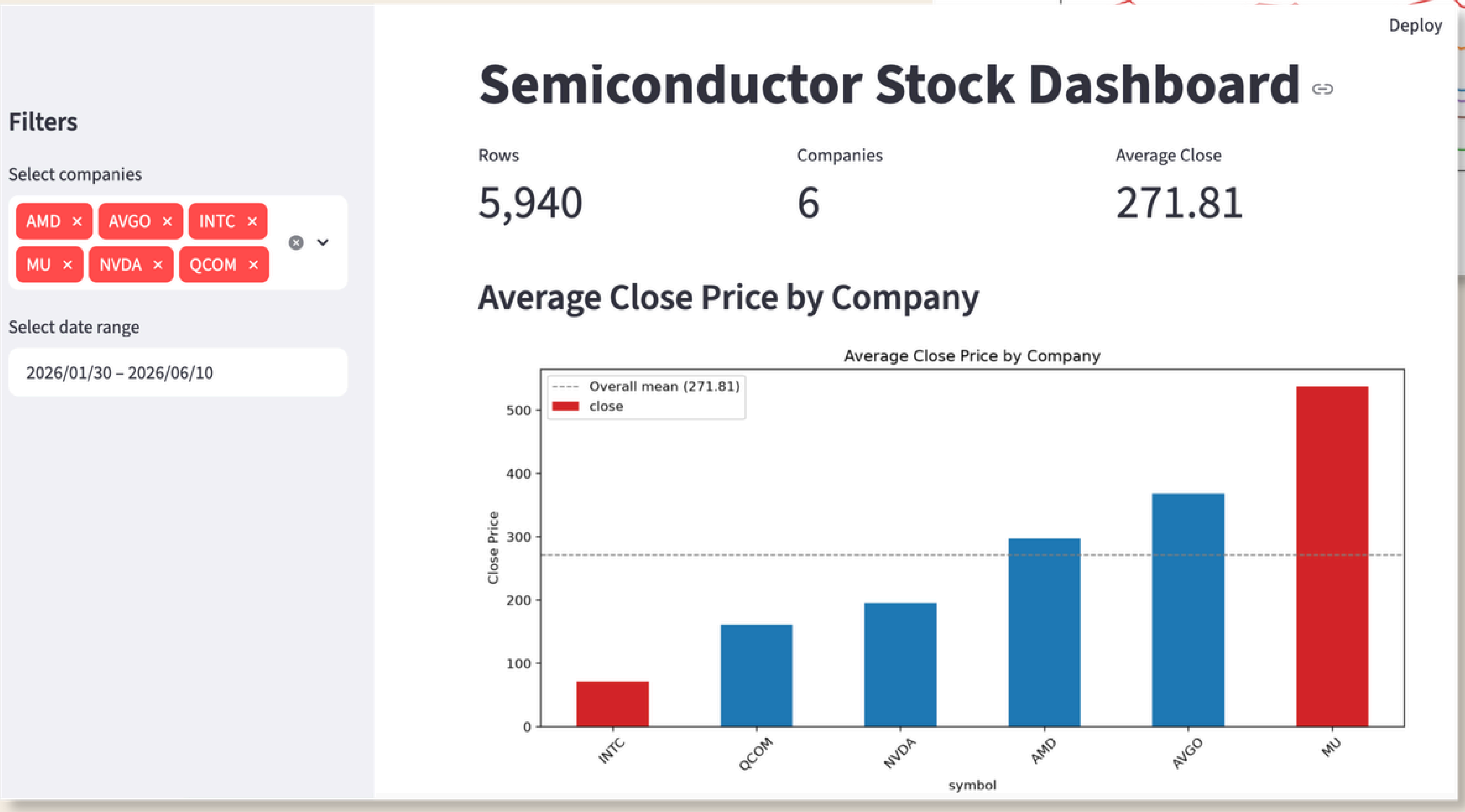
MSE 108.7496

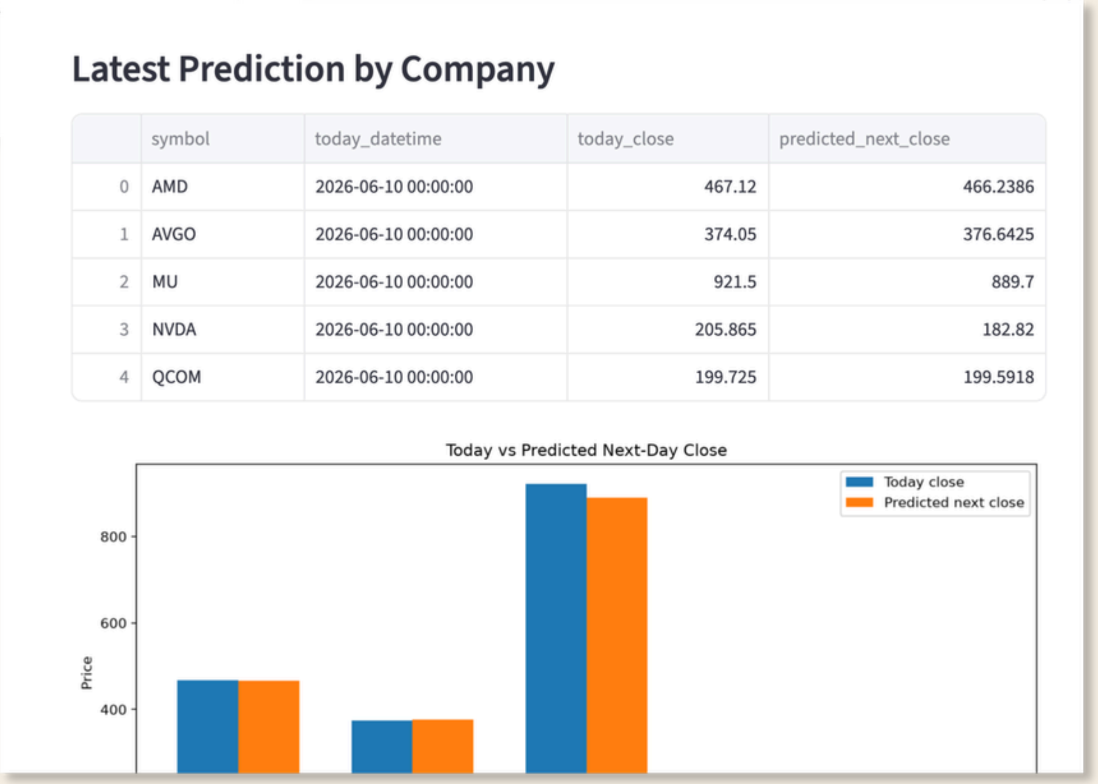
R^2 0.9969



06. STREAMLIT DASHBOARD







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1	AVGO	2026-06-10 00:00:00	374.05	376.6425
2	MU	2026-06-10 00:00:00	921.5	889.7
3	NVDA	2026-06-10 00:00:00	205.865	182.82
4	QCOM	2026-06-10 00:00:00	199.725	199.5918

07. LINK

github: https://github.com/leesujin113/Final-Term/tree/main

08. AI USAGE

Images in slide 4,5 are created by AI

23.06.2026

THANK YOU